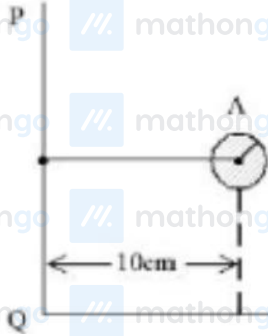


Q1 - 24 January - Shift 1

Solid sphere A is rotating about an axis PQ. If the radius of the sphere is 5 cm then its radius of gyration about PQ will be $\sqrt{x}$ cm. The value of x is _____.



Space for your notes:

Q2 - 24 January - Shift 2

A uniform solid cylinder with radius R and length L has moment of inertia I_1 , about the axis of cylinder. A concentric solid cylinder of radius $R' = \frac{R}{2}$ and length $L' = \frac{L}{2}$ is carved out of the original cylinder. If I_2 is the moment of inertia of the carved out portion of the cylinder then $\frac{I_1}{I_2} =$ _____

Space for your notes:

(Both I_1 and I_2 are about the axis of the cylinder)

Q3 - 25 January - Shift 1

A car is moving with a constant speed of 20 m/s in a circular horizontal track of radius 40 m. A bob is suspended from the roof of the car by a massless string. The angle made by the string with the vertical will be : (Take $g = 10 \text{ m/s}^2$)

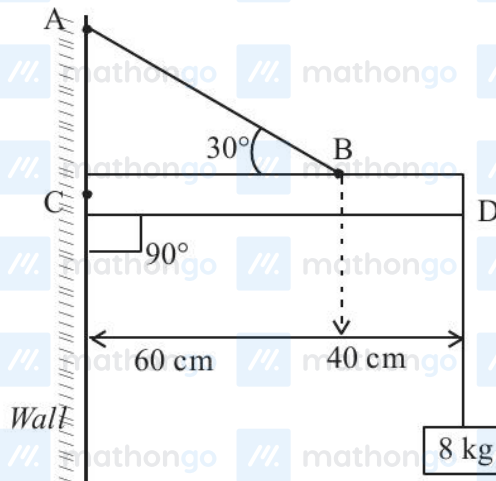
(1) $\frac{\pi}{6}$
(3) $\frac{\pi}{4}$

(2) $\frac{\pi}{2}$
(4) $\frac{\pi}{3}$

Space for your notes:

Q4 - 25 January - Shift 1

An object of mass 8 kg is hanging from one end of a uniform rod CD of mass 2 kg and length 1 m pivoted at its end C on a vertical wall as shown in figure. It is supported by a cable AB such that the system is in equilibrium. The tension in the cable is : (Take $g = 10 \text{ m/s}^2$)



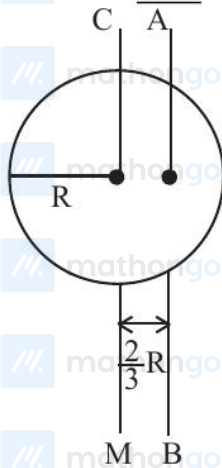
(1) 240 N
(3) 300 N

(2) 90 N
(4) 30 N

Space for your notes:

Q5 - 25 January - Shift 1

I_{CM} is moment of inertia of a circular disc about an axis (CM) passing through its center and perpendicular to the plane of disc. I_{AB} is its moment of inertia about an axis AB perpendicular to plane and parallel to axis CM at a distance $\frac{2}{3}R$ from center. Where R is the radius of the disc. The ratio of I_{AB} and I_{CM} is $x:9$. The value of x is _____.



Space for your notes:

Q6 - 25 January - Shift 2

If a solid sphere of mass 5 kg and a disc of mass 4 kg have the same radius. Then the ratio of moment of inertia of the disc about a tangent in its plane to the moment of inertia of the sphere about its tangent will be $\frac{x}{7}$. The value of x is _____.

Space for your notes:

Q7 - 29 January - Shift 1

A car is moving on a horizontal curved road with radius 50 m. The approximate maximum speed of car will be, if friction between tyres and road is 0.34. [Take $g = 10 \text{ ms}^{-2}$]

- (1) 3.4 ms^{-1} (2) 22.4 ms^{-1}
(3) 13 ms^{-1} (4) 17 ms^{-1}

Space for your notes:

Q8 - 29 January - Shift 1

A solid sphere of mass 2kg is making pure rolling on a horizontal surface with kinetic energy 2240 J.

The velocity of centre of mass of the sphere will be _____ ms^{-1} .

Space for your notes:

Q9 - 29 January - Shift 2

An object moves at a constant speed along a circular path in a horizontal plane with centre at the origin. When the object is at $x = +2\text{m}$, its velocity is $-4\hat{j} \text{ m/s}$. The object's velocity (v) and acceleration (a) at $x = -2 \text{ m}$ will be

- (1) $v = 4\hat{i} \text{ m/s}$, $a = 8\hat{j} \text{ m/s}^2$
(2) $v = 4\hat{j} \text{ m/s}$, $a = 8\hat{i} \text{ m/s}^2$
(3) $v = -4\hat{j} \text{ m/s}$, $a = 8\hat{i} \text{ m/s}^2$
(4) $v = -4\hat{i} \text{ m/s}$, $a = -8\hat{j} \text{ m/s}^2$

Space for your notes:

Q10 - 29 January - Shift 2

A car is moving on a circular path of radius 600 m such that the magnitudes of the tangential acceleration and centripetal acceleration are equal.

The time taken by the car to complete first quarter of revolution, if it is moving with an initial speed of 54 km/hr is $t(1 - e^{-\pi/2})$ s. The value of t is _____.

Space for your notes:

Q11 - 30 January - Shift 1

A thin uniform rod of length 2m, cross sectional area 'A' and density 'd' is rotated about an axis passing through the centre and perpendicular to its length with angular velocity ω . If value of ω in terms of its rotational kinetic energy E is $\sqrt{\frac{\alpha E}{Ad}}$ then the value of α is _____.

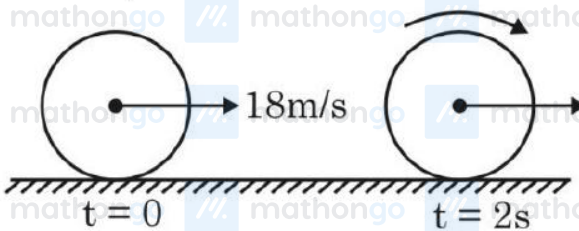
Space for your notes:

Q12 - 30 January - Shift 2

A uniform disc of mass 0.5 kg and radius r is projected with velocity 18 m/s at $t = 0$ s on a rough horizontal surface. It starts off with a purely sliding motion at $t = 0$ s. After 2s it acquires a purely rolling motion (see figure). The total kinetic energy of the disc after 2s will be _____ J

Space for your notes:

(given, coefficient of friction is 0.3 and $g = 10 \text{ m/s}^2$).



Q13 - 31 January - Shift 2

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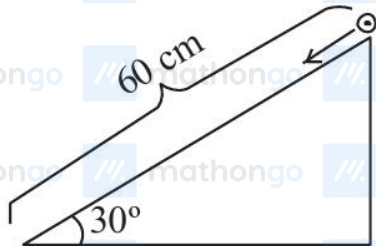
Two discs of same mass and different radii are made of different materials such that their thicknesses are 1 cm and 0.5 cm respectively. The densities of materials are in the ratio 3:5. The moment of inertia of these discs respectively about their diameters will be in the ratio of $\frac{x}{6}$. The value of x is _____.

Space for your notes:

Q14 - 01 February - Shift 1

A solid cylinder is released from rest from the top of an inclined plane of inclination 30° and length 60 cm. If the cylinder rolls without slipping, its speed upon reaching the bottom of the inclined plane is _____ ms^{-1} . (Given $g = 10 \text{ ms}^{-2}$)

Space for your notes:



Q15 - 01 February - Shift 2

Moment of inertia of a disc of mass M and radius 'R' about any of its diameter is $\frac{MR^2}{4}$. The moment of inertia of this disc about an axis normal to the disc and passing through a point on its edge will be, $\frac{x}{2} MR^2$. The value of x is _____.

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (110)

Q2 (32)

Q3 (3)

Q4 (3)

Q5 (17)

Q6 (5)

Q7 (3)

Q8 (40)

Q9 (2)

Q10 (40)

Q11 (3)

Q12 (54)

Q13 (5)

Q14 (2)

Q15 (3)

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To practice more chapter-wise JEE Main PYQs, [click here to download the MARKS app](#) from Playstore

Q1 (110)

$$I_{cm} = \frac{2}{5}MR^2$$

$$I_{PQ} = I_{cm} + md^2$$

$$I_{PQ} = \frac{2}{5}mR^2 + m(10\text{cm})^2$$

For radius of gyration

$$I_{PQ} = mk^2$$

$$k^2 = \frac{2}{5}R^2 + (10\text{cm})^2$$

$$= \frac{2}{5}(5)^2 + 100$$

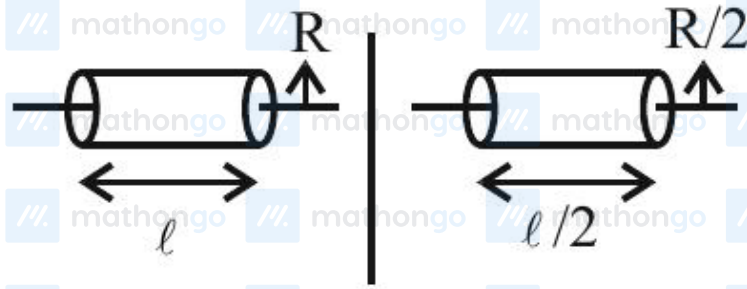
$$= 10 + 100 = 110$$

$$k = \sqrt{110}\text{cm}$$

$$x = 110$$

Q2 (32)

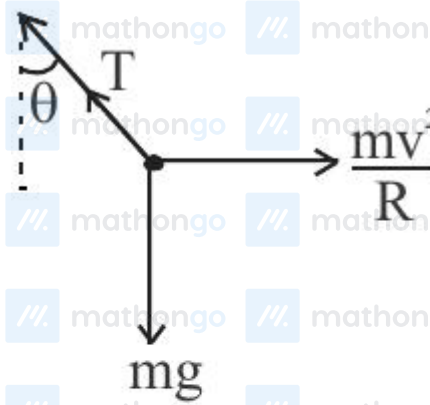
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$$I_1 = \frac{m_1 R^2}{2} \quad I_2 = \frac{m_2 (R/2)^2}{2}$$

$$\frac{I_1}{I_2} = \frac{4m_1}{m_2} = \frac{4 \cdot \rho \pi R^2 l}{\rho \cdot \frac{\pi R^2}{4} \times \frac{l}{2}} \Rightarrow \frac{I_1}{I_2} = 32$$

Q3 (3)



$$T \cos \theta = mg$$

$$T \sin \theta = \frac{mv^2}{R}$$

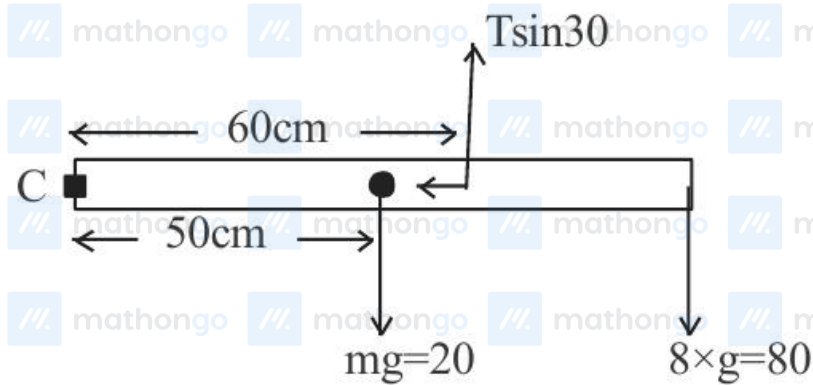
$$\tan \theta = \frac{v^2}{Rg}$$

$$\tan \theta = \frac{20^2}{40 \times 10}$$

$$\tan \theta = 1$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

Q4 (3)



Taking torque about point C

$$\frac{T}{2} \times 60 = 20 \times 50 + 80 \times 100$$

$$\Rightarrow 3T = 100 + 800$$

$$\Rightarrow T = 300\text{ N}$$

Q5 (17)

$$I_{\text{cm}} = \frac{mR^2}{2}$$

$$I_{\text{AB}} = \frac{mR^2}{2} + m\left(\frac{2R}{3}\right)^2 = \frac{17}{18}mR^2$$

$$\frac{I_{\text{AB}}}{I_{\text{cm}}} = \frac{17}{9} \Rightarrow x = 17$$

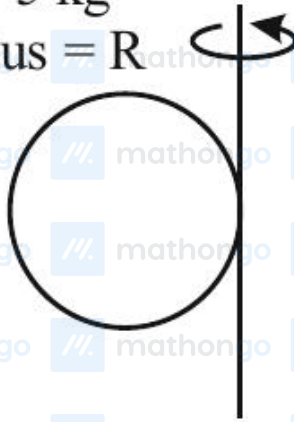
Q6 (5)

tangent will be $\frac{\omega}{7}$. The value of x is

$$m_1 = 5 \text{ kg}$$

$$\text{Radius} = R$$

Solid
sphere



$$I_1 = \frac{2}{5} m_1 R^2 + m_1 R^2$$

$$I_1 = m_1 R^2 \left(\frac{7}{5} \right)$$

$$I_1 = 7R^2$$

$$m_2 = 4 \text{ kg}$$

$$\text{Radius} = R$$

Disc



$$I_2 = \frac{m_2 R^2}{4} + m_2 R^2$$

$$I_2 = \frac{5}{4} m_2 R^2$$

$$I_2 = 5R^2$$

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Q7 (3)

$$f_s = \frac{mv^2}{r}$$

For maximum speed in safe turning,

$$f_s = f_s \text{ max} = \mu mg$$

$$v_{\text{max}} \text{ (for safe turning)} = \sqrt{\mu rg}$$

$$= \sqrt{0.34 \times 50 \times 10} \approx 13 \text{ m/s}$$

Q8 (40)

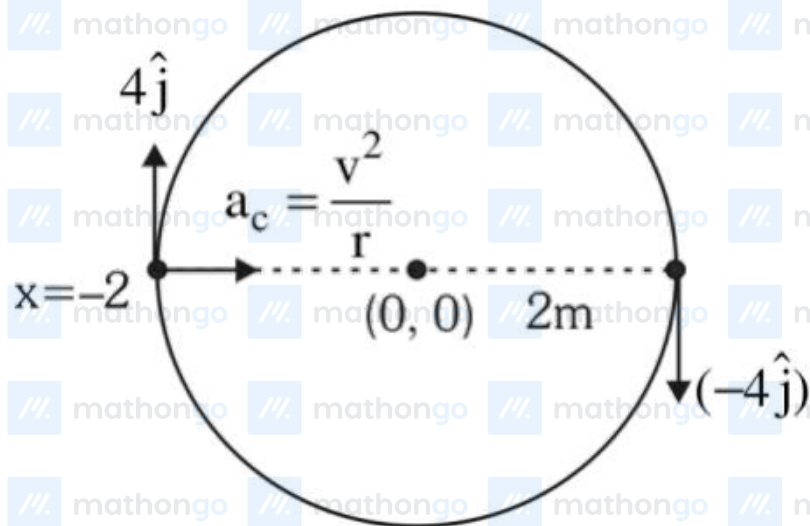
$$KE = \frac{1}{2} mv^2 + \frac{1}{2} I\omega^2$$

$$2240 = \frac{1}{2} 2(v)^2 + \frac{1}{2} \frac{2}{5} (2) R^2 \cdot \left(\frac{v}{R}\right)^2$$

$$2240 = v^2 + \frac{2}{5} v^2$$

$$\Rightarrow v = 40 \text{ m/s}$$

Q9 (2)



$$a_c = \frac{V^2}{r} = \frac{4^2}{2} = \frac{16}{2} = 8 \text{ m/s}^2$$

$$\vec{V} = 4\hat{j}$$

$$\vec{a}_c = 8\hat{i}$$

Q10 (40)

$$v \frac{dv}{dx} = \frac{v^2}{R} \Rightarrow \int_{15}^v \frac{dv}{v} = \frac{1}{R} \int_0^x dx$$

$$v = 15e^{x/R}$$

$$\frac{dx}{dt} = 15e^{x/R}$$

$$\frac{\pi R}{2} \int_0^{\pi/2} e^{-x/R} dx = 15 \int_0^{t_0} dt$$

$$t_0 = 40 \left(1 - e^{-\pi/2} \right)$$

Q11 (3)

$$(KE)_{\text{Rotational}} = \frac{1}{2} I \omega^2 = E$$

$$E = \frac{1}{2} \frac{m \ell^2}{12} \omega^2$$

$$E = \frac{1}{2} \frac{dA \ell^3}{12} \omega^2$$

$$E = \frac{dA (2)^3}{24} \omega^2$$

$$\sqrt{\frac{3E}{dA}} = \omega$$

$$\alpha = 3 \text{ Ans.}$$

Q12 (54)

$$a = -\mu_k g = -3$$

$$V = 18 - 3 \times 2$$

$$V = 12 \text{ m/s}$$

$$KE = \frac{1}{2}mv^2 + \frac{1}{2}\frac{mr^2}{r^2}v^2$$

$$KE = \frac{3}{4}mv^2$$

$$KE = 3 \times 18 = 54 \text{ J}$$

Q13 (5)

No Solution

Q14 (2)

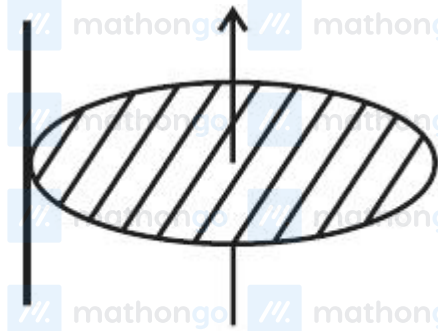
$$v = \sqrt{\frac{2gh}{1 + \frac{k^2}{R^2}}}$$

Where $h = 60 \sin 30^\circ = 30 \text{ cm}$

$$k^2 = \frac{R^2}{2}$$

Q15 (3)

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$$I = I_{\text{cm}} + Md^2$$

$$= \frac{MR^2}{2} + MR^2$$

$$= \frac{3}{2}MR^2$$